Bing's Theorem 3 (1948), as it was formalized: a human-readable account of the Lean proof

Abstract

This note states, in ordinary mathematical language, the theorem that the Lean development in this repository proves under the name `HPC.Fold.theorem3` (file `Fold/Open.lean`), and gives the proof that the Lean code actually checks. The formal proof does not follow Bing's original argument. It first reduces the statement about chains to a statement about one integer sequence, the *minimal-index sequence*. It then proves that statement with an explicit construction, a *greedy jump zigzag*. Crookedness is used in exactly two small lemmas. Section 8 matches each step to the Lean declaration that proves it.

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1 The statement

Throughout, X is a metric space. (Only its topology matters, and only in that links are open sets.) We write $[n] = \{0, 1, \ldots, n-1\}$.

Definition 1.1 (Chain)**.** A *chain of length* n in X is a family $E = (E_0, \ldots, E_{n-1})$ of open subsets of X, its *links*, such that for all $i, j \in [n]$

$$E_i \cap E_j \neq \emptyset \iff |i - j| \leq 1.$$

(Taking $i = j$ shows that every link is nonempty. Links need not be connected.)

Definition 1.2 (Refinement, 1948 crookedness)**.** Let E be a chain of length n and D a chain of length k.

(a) *E refines D* if every link of E is contained in some link of D.

(b) E is *crooked in D* (in the 1948 sense) if E refines D and the following holds. Let $h, k' \in [k]$ with $k' > h+2$, and let $i, j \in [n]$ with $E_i \cap D_h \neq \emptyset$ and $E_j \cap D_{k'} \neq \emptyset$. Then there are indices r, s strictly between i and j, with r nearer to i (that is, $i < r < s < j$ or $i > r > s > j$), such that

$$E_r \subseteq D_{k'-1} \qquad \text{and} \qquad E_s \subseteq D_{h+1}.$$

The hypothesis asks that E_i and E_j *meet* D_h and $D_{k'}$. This is what distinguishes the 1948 definition from the 1951 one, where they must be *contained in* D_h and $D_{k'}$. The Lean name of this notion is `Chain.IsCrookedIn'`.

Definition 1.3 (Block maps and block consolidations)**.** A *block map* is a surjection $\beta : [n] \to [q]$ with $|\beta(t+1) - \beta(t)| \leq 1$ for all t. Its *blocks* are the fibres $\beta^{-1}(u)$, $u \in [q]$. Given a chain E of length n, the *block consolidation* E^β is the family

$$E^\beta_u \;=\; \bigcup_{t:\ \beta(t)=u} E_t \qquad (u \in [q]).$$

The Lean development proves that E^β is again a chain (`Chain.ofBlocks`, `blockLink_adjacent`). A block need not be a set of consecutive indices. For example, $\beta = (0, 1, 0)$ puts E_0 and E_2 into the same link. So E^β is a consolidation of E that may "fold" E.

For a chain D of length k and $j \in \mathbb{N}$ we write $D_{[j,j+3)} = \bigcup\{D_{j'} : j \leq j' < j+3,\ j' < k\}$ for the union of (at most) three consecutive links (Lean: `D.unionRange j 3`).

Theorem 1.4 (Bing 1948, Theorem 3: block reading, three-link form; Lean: `HPC.Fold.theorem3`)**.** *Let E be a chain of length $n > 0$ and D a chain of length k in a metric space X, and let E be crooked in D in the 1948 sense. Then for every $i \in [n]$ there is a block map $\beta : [n] \to [q]$ such that*

(i) $\beta(i) = 0$, so the link E_i lies in the first link of E^β;

(ii) for every $u \in [q]$ there is $j \in \mathbb{N}$ with $E^\beta_u \subseteq D_{[j,j+3)}$. That is, every link of the consolidation lies in three consecutive links of D.

The Lean proof uses only the standard axioms (`propext`, `Classical.choice`, `Quot.sound`). The recorded output is in `axioms-batch77.txt`.

Plan of the proof.

1. Section 2 reduces the theorem to one integer sequence. Read E through its minimal-index sequence $\rho(t) = \min\{j : E_t \subseteq D_j\}$. It then suffices that the two *arms* of ρ from i (forwards and backwards) *fold onto a common walk* (Definition 2.3). This is the statement called `S3`.

2. Section 3 extracts the only consequence of crookedness used later: a "level crookedness" property of ρ (Lemma 3.1).

3. Sections 4–7 prove `S3`. The common walk is a *jump zigzag*. Its legs skip the two levels next to each turning point, so folding onto it becomes a condition on the order of events with no timing constraint (Section 4). The turning levels are chosen greedily (Section 5). Two short crookedness lemmas show that the greedy choice never gets stuck (Section 6). Each phase widens the zigzag and the sequence is bounded, so the construction terminates (Section 7).

2 Reduction to an integer sequence

Fix E crooked in D, so in particular E refines D.

Definition 2.1 (Minimal and maximal index)**.** For $t \in [n]$ let $\rho(t)$ be the least and $\sigma(t)$ the greatest index j with $E_t \subseteq D_j$ (Lean: `Refines.minIdx`, `Refines.maxIdx`).

Lemma 2.2. *(a) If $E_t \subseteq D_j$ then $\rho(t) \le j \le \rho(t)+1$. In particular $\rho(t) \le \sigma(t) \le \rho(t)+1$.*

(b) ρ is 1-Lipschitz: $|\rho(t+1) - \rho(t)| \le 1$.

Proof. (a) E_t is nonempty and lies in both $D_{\rho(t)}$ and D_j. So these two links meet, and $|j-\rho(t)| \le 1$. Minimality gives $\rho(t) \le j$. (b) $E_t \cap E_{t+1} \neq \emptyset$, so $D_{\rho(t)} \cap D_{\rho(t+1)} \neq \emptyset$. □

Definition 2.3 (Arms and folds)**.** Let $f : [m] \to \mathbb{Z}$, $\ell \in \mathbb{N}$ and $c : \mathbb{N} \to \mathbb{Z}$ be arbitrary. We say that f *folds onto the walk c of length ℓ* if there is a *time assignment* $\beta : [m] \to \mathbb{N}$ with

(F1) $\beta(j) < \ell$ for all j;

(F2) $|\beta(j+1) - \beta(j)| \le 1$ for all j;

(F3) $\beta(0) = 0$;

(F4) $|f(j) - c(\beta(j))| \le 1$ for all j.

No condition at all is imposed on c. In particular consecutive values of c may differ by more than one, and this freedom is essential below. (Lean: `FoldsOnto`.) For $\rho : [n] \to \mathbb{Z}$ and $i \in [n]$, the *forward arm* is $j \mapsto \rho(i+j)$ on $[n-i]$ and the *backward arm* is $j \mapsto \rho(i-j)$ on $[i+1]$ (Lean: `armG`, `armT`).

Proposition 2.4 (From a common fold to Theorem 1.4; Lean: `exists_blocks_first_of_folds`)**.** *Suppose both arms of ρ from i fold onto the same walk c of length ℓ. Then the conclusion of Theorem 1.4 holds for i.*

Proof. Let β_G, β_T be the two time assignments. Define $\beta : [n] \to \mathbb{N}$ by $\beta(t) = \beta_G(t-i)$ for $t \ge i$ and $\beta(t) = \beta_T(i-t)$ for $t \le i$. The two definitions agree at $t = i$, where both give 0. Then β is 1-Lipschitz, $\beta(i) = 0$, and $|\rho(t) - c(\beta(t))| \le 1$ for all t. By the discrete intermediate value theorem, β maps onto $\{0, \dots, \max \beta\}$, so it is a block map onto $[q]$ with $q = \max\beta + 1$. If $\beta(t) = \beta(t') = u$, then $\rho(t)$ and $\rho(t')$ both lie in $\{c(u)-1, c(u), c(u)+1\}$. Let j be the least value of ρ on the block $\beta^{-1}(u)$. Every t in the block has $\rho(t) \in \{j, j+1, j+2\}$ and $E_t \subseteq D_{\rho(t)}$. Hence $E^{\beta}_u \subseteq D_{[j,j+3)}$. □

Let $L = \min \rho$ and $H = \max \rho$.

- *Degenerate case $H \le L+2$* (Lean: `exists_blocks_first_of_range_le_two`). Take the one-block map $\beta \equiv 0$. Every E_t lies in $D_{\rho(t)} \subseteq D_{[L,L+3)}$.
- *Main case $H \ge L+3$.* By Proposition 2.4 it is enough to know that both arms fold onto a common walk. This is the statement `S3` below, applied to ρ and σ. The pair (ρ, σ) satisfies the hypotheses of `S3` by Lemma 2.2 and by the "faithful model" lemma `isCrookedIvF_minIdx_maxIdx`. The proof of `S3` uses only Lemma 3.1 below, which is derived from that lemma. This combination is the Lean theorem `theorem3_of_S3`.

Theorem 2.5 (`S3`; proved as `HPC.Fold.Z77.s3_zig` and restated as `HPC.Fold.s3_general`)**.** *Let $\rho, \sigma : [n] \to \mathbb{Z}$, $i \in [n]$ and $L, H \in \mathbb{Z}$. Suppose that ρ is 1-Lipschitz, that (ρ, σ) lies in the faithful crooked model (`IsCrookedIvF`, recalled in Section 3), and that $L \le \rho \le H$. Then there are $\ell \in \mathbb{N}$ and $c : \mathbb{N} \to \mathbb{Z}$ such that both arms of ρ from i fold onto c of length ℓ.*

Remark 2.6. As stated in Lean, `S3` carries three more hypotheses: $L + 3 \le H$, and that L and H are attained by ρ. They are left over from earlier approaches, and the final proof does not use them. The bounds $L \le \rho \le H$ are used only to make the construction terminate. The auxiliary sequence σ enters only through Lemma 3.1.

3 The only consequence of crookedness that is used

The faithful model `IsCrookedIvF`(ρ, σ) consists of four conditions: $\rho \le \sigma \le \rho+1$; an adjacency condition on neighbouring intervals; and a crookedness clause stated for the integer intervals $[\rho(t), \sigma(t)]$. The Lean lemma `isCrookedIvF_minIdx_maxIdx` shows that the minimal and maximal index sequences of a 1948-crooked chain satisfy it. The proof of `S3` uses only the following consequence. We state it directly for the chain.

Lemma 3.1 (Level crookedness; Lean: `crkOn_of_isCrookedIvF`, predicate `CrkOn`)**.** *Let $s < t$ in $[n]$.*

(a) If $\rho(t) \ge \rho(s) + 3$, there are $s < r < r' < t$ with $\rho(r) \in \{\rho(t) - 2, \rho(t) - 1\}$ and $\rho(r') \in \{\rho(s), \rho(s) + 1\}$.

(b) If $\rho(s) \ge \rho(t) + 3$, there are $s < r < r' < t$ with $\rho(r) \in \{\rho(t), \rho(t) + 1\}$ and $\rho(r') \in \{\rho(s) - 2, \rho(s) - 1\}$.

In words: between two times whose levels differ by at least three, the sequence first comes within one of the later *level, and after that within one of the* earlier *level.*

Proof. (a) Put $h = \rho(s)$ and $k' = \rho(t)$, so $k' > h + 2$. Since $E_s \subseteq D_h$ and $E_t \subseteq D_{k'}$, the link E_s meets D_h and E_t meets $D_{k'}$. Crookedness with $i = s$, $j = t$ gives $s < r < r' < t$ (the order is forced by $s < t$) with $E_r \subseteq D_{k'-1}$ and $E_{r'} \subseteq D_{h+1}$. By Lemma 2.2(a), $\rho(r) \in \{k' - 2, k' - 1\}$ and $\rho(r') \in \{h, h + 1\}$.

(b) Apply crookedness with $h = \rho(t)$, $k' = \rho(s)$, $i = t$, $j = s$. This gives $t > r_1 > r_2 > s$ with $E_{r_1} \subseteq D_{\rho(s)-1}$ and $E_{r_2} \subseteq D_{\rho(t)+1}$. Take $r = r_2$ and $r' = r_1$ and use Lemma 2.2(a). □

The Lean proof goes through `IsCrookedIvF` and specializes its crookedness clause to $h = \rho(s)$, $k = \rho(t)$, with the witnesses $u = s$ and $u = t$. The width condition $\sigma \le \rho + 1$ then gives the same two-element ranges.

Notation for the rest of the proof. Fix ρ, i, $L \le \rho \le H$. Re-index at the base point: $S(x) = \rho(i + x)$. The two arms are $f_\uparrow(j) = S(j)$ for $j < m_\uparrow = n - i$ and $f_\downarrow(j) = S(-j)$ for $j < m_\downarrow = i + 1$. Both are 1-Lipschitz, take values in $[L, H]$, and start at $p = S(0) = \rho(i)$. Lemma 3.1 holds for S on its whole domain, and also for $x \mapsto S(-x)$, since reading the sequence backwards swaps the two clauses (`CrkOn.reverse`). This means the two arms play symmetric roles. In the arguments below one arm is called the *left* arm and the other the *right* arm. A time j on the left arm is the index $-j$ of S, and a time j on the right arm is the index $+j$. So every left-arm time comes before every right-arm time in the order used by Lemma 3.1.

4 Jump zigzags and the fitting condition

Definition 4.1 (Jump walk; Lean: `jwalk`, `jlen`)**.** Let $T_0 \in \mathbb{Z}$ and a list of turning levels $T_1, \dots, T_K$ with $|T_r - T_{r-1}| \geq 3$ be given. For a leg $a \to b$ put $\lambda = |b - a| - 2 \geq 1$. The leg is traversed as

$$a,\ a \pm 3,\ a \pm 4,\ \dots,\ b \mp 1 \qquad (\lambda \text{ values}),$$

with $+$ if $a < b$ and $-$ if $a > b$. The next leg then starts at b. After the last leg the walk stays at T_K. Its length is $\sum \lambda + 1$. The walk thus jumps from each turning level straight to three levels beyond it, skipping the two levels in between.

Definition 4.2 (Envelope, arrival level, fitting; Lean: `Env`, `arr`, `Fit`)**.** For a leg $a \to b$:

- the *envelope* is $\mathrm{Env}(a, b) = [\min(a, b) - 1,\ \max(a, b) + 1]$;
- the *arrival level* is $\mathrm{arr}(a, b) = b - 1$ if $a < b$ and $b + 1$ if $a > b$.

Let f be defined on $[m]$. The relation $\mathrm{Fit}(f, \tau, a, [T_1, \dots, T_K])$ ("from time τ, f fits the legs $a \to T_1 \to \cdots \to T_K$") is defined by recursion on the list:

- it is false for the empty list;
- for $[b, \mathrm{rest}]$ it holds iff $|b - a| \geq 3$ and one of the following:
  - (i) $f(j) \in \mathrm{Env}(a, b)$ for all $j \in [\tau, m)$ (the arm stays on this leg for good); or
  - (ii) there is $\tau' \in [\tau, m)$ with $f(\tau') = \mathrm{arr}(a, b)$, $f(j) \in \mathrm{Env}(a, b)$ for all $j \in [\tau, \tau']$, and $\mathrm{Fit}(f, \tau', b, \mathrm{rest})$.

Lemma 4.3 (Fitting gives a fold; Lean: `fold_of_fit`)**.** *Let f be 1-Lipschitz on $[m]$. Suppose $\mathrm{Fit}(f, \tau, a, ts)$ and $|f(\tau) - a| \leq 1$. Then there is a 1-Lipschitz β on $[\tau, m)$ with $\beta(\tau) = 0$, values less than the length of the jump walk c of (a, ts), and $|f(j) - c(\beta(j))| \leq 1$ for all $j \in [\tau, m)$.*

Proof. Consider a leg $a \to b$ with $a < b$; the case $a > b$ is symmetric. For a level x define the *position* $\mathrm{pos}(x) = \min(\max(x - a - 1, 0), \lambda)$. Position $u < \lambda$ has walk value a when $u = 0$ and $a + 2 + u$ otherwise, and position λ is the start b of the next leg. One checks that for every $x \in \mathrm{Env}(a, b) = [a - 1, b + 1]$ the value at $\mathrm{pos}(x)$ is within one of x:

- if $x \leq a + 1$, the position is 0 and the value is a;
- if $x = a + 1 + u$ with $1 \leq u < \lambda$, the value is $a + 2 + u = x + 1$;
- if $x \geq b - 1$, the position is λ and the value is b.

While the arm is on the leg, set $\beta(j) = (\text{offset of the leg}) + \mathrm{pos}(f(j))$. Since pos is 1-Lipschitz and f is 1-Lipschitz, so is β. At the switching time τ' we have $f(\tau') = b - 1$, so $\mathrm{pos}(f(\tau')) = \lambda$. This is position 0 of the next leg, because $|f(\tau') - b| \leq 1$. So the time assignments of consecutive legs agree at τ'. Induction on the list finishes the proof. □

Remark 4.4. This is the point of the jumps. On each leg the time is a function of the current *level* alone. If the walk moved one level per step, an arm would also have to meet timing constraints: it would have to wait at turning points for the walk. With the jumps, whether an arm folds depends only on the order in which it visits levels.

5 One greedy phase

The construction alternates *up phases* and *down phases*. Each arm is either *finished* or *live*, and a live arm has a current *start time*. We state the up phase. The down phase is the same statement for the negated arms, with levels reflected (Lean: `down_step` is derived from `up_step`).

Lemma 5.1 (One up phase; Lean: `up_step`)**.** *Let $b+2 \le a$. For each arm $e \in \{\uparrow, \downarrow\}$ let τ_e be an optional start time with $|f_e(\tau_e) - b| \le 1$. Assume the up phase is* not stuck*:*

> ($\mathrm{N}_\uparrow$) *there is no live arm e and time $d \ge \tau_e$ with $f_e(d) = b-2$ and $f_e(j) \le a-1$ for all $j \in [\tau_e, d)$.*

Then there are a level $t \ge a+1$ and optional new start times σ_e (defined only for arms that were live) with the following properties.

(1) If e is live and σ_e is undefined, then $b-1 \le f_e(j) \le t-2$ for all $j \ge \tau_e$.

(2) If σ_e is defined, then $\tau_e < \sigma_e$, $f_e(\sigma_e) = t-1$, and $b-1 \le f_e(j) \le t-2$ for $j \in [\tau_e, \sigma_e)$.

(3) Either no σ_e is defined, or $t \le H+1$ and some arm e with σ_e defined later drops: there is $d > \sigma_e$ with $f_e(d) = b-2$ and $b-1 \le f_e(j) \le t-1$ for $j \in [\tau_e, d)$.

Proof. For a live arm e, suppose it drops below $b-1$ after τ_e. Its first such time d_e has $f_e(d_e) = b-2$, by the Lipschitz condition. Let M_e be the maximum of f_e on $[\tau_e, d_e)$. By ($\mathrm{N}_\uparrow$), $M_e \ge a$. Put $v_e = M_e + 1$. Otherwise, or if e is finished, put $v_e = \max(a+1, H+2)$. Let $t = \min_e v_e \ge a+1$. Every drop of a live arm below $b-1$ is preceded by a visit to level $\ge v_e - 1 \ge t-1$. So each live arm either stays in $[b-1, t-2]$ forever, or reaches $t-1$ at a first time σ_e while staying in $[b-1, t-2]$ before it. This gives (1) and (2).

For (3), first suppose $t = \max(a+1, H+2)$. Then $t-1 \ge H+1 > H$, so no arm reaches $t-1$, and no σ_e is defined. Otherwise $t = v_e = M_e + 1 \le H+1$ for an arm e that drops at d_e after reaching $M_e = t-1$. That arm cannot be finished, by (1). So $\sigma_e < d_e$, and the bounds in (3) follow from the definition of M_e. □

A leg $b \to t$ is fitted by every live arm. By (1) a finished arm stays in $[b-1, t-2] \subseteq \mathrm{Env}(b, t)$. By (2) a switching arm stays in the envelope up to σ_e, where it is at $t-1 = \mathrm{arr}(b, t)$. The leg has height $t - b \ge 3$. Neither part uses crookedness.

6 Why the greedy choice never gets stuck: two crookedness lemmas

The two lemmas below are all the crookedness the construction uses. In both, $b \le a$, and the left and right arms are as described at the end of Section 3.

Lemma 6.1 (Up core; Lean: `up_core`)**.** *The following configuration is impossible:*

- *left arm: $\ge b$ before time τ, $\le a-1$ on $[\tau, d_L)$, and $= b-2$ at d_L;*
- *right arm: $\ge b$ before time d_R, and $= a+2$ at d_R.*

Proof. Apply Lemma 3.1(a) to the index $-d_L$ (level $b-2$) and the index d_R (level $a+2$). The difference is $a - b + 4 \ge 3$. We get indices $-d_L < r < r' < d_R$ with $S(r) \in \{a, a+1\}$ and $S(r') \in \{b-2, b-1\}$.

- If $r' \geq 0$, then r' is a right-arm time before d_R, so $S(r') \geq b$. Contradiction.
- If $r' < 0$, then $-r'$ is a left-arm time with level $< b$, so $-r' \geq \tau$. Then $-r$ is a left-arm time with $\tau \leq -r' < -r < d_L$, so $S(r) \leq a - 1$. Contradiction. □

Lemma 6.2 (Down core; Lean: `down_core`)**.** *The following configuration is impossible:*

- *left arm: $\geq b$ before τ, $\leq a - 2$ on $[\tau, \sigma)$, $\geq b + 1$ on $[\sigma, rise)$, and $= a + 2$ at rise;*
- *right arm: $\geq b$ before τ', $\leq a - 1$ on $[\tau', d)$, and $= b - 2$ at d.*

Proof. Apply Lemma 3.1(b) to the index $-rise$ (level $a + 2$) and the index d (level $b - 2$). We get $-rise < r < r' < d$ with $S(r) \in \{b - 2, b - 1\}$ and $S(r') \in \{a, a + 1\}$.

- If $r \geq 0$: when $r < \tau'$ the right arm has $S(r) \geq b$, a contradiction. When $r \geq \tau'$, the index r' is a right-arm time in $[\tau', d)$ with $S(r') \geq a$, again a contradiction.
- If $r < 0$: put $u = -r$, a left-arm time with level $\leq b - 1$. Then $\tau \leq u < \sigma$. Now apply Lemma 3.1(b) a second time, to $-rise$ (level $a + 2$) and r (level $\leq b - 1$). We get $-rise < q < q' < r$ with $S(q) \leq S(r) + 1 \leq b$ and $S(q') \geq a$. The left-arm time $-q'$ lies in $(u, rise)$ and has level $\geq a > a - 2$, so $-q' \geq \sigma$. Then $-q \in (-q', rise) \subseteq [\sigma, rise)$, so $S(q) \geq b + 1$. Contradiction. □

From these we get the three "not stuck" statements.

Lemma 6.3 (Lean: `first_direction`)**.** *With $p = S(0)$, at least one of the following holds: (A) no arm reaches $p - 3$ while staying $\leq p$ before; or (B) no arm reaches $p + 3$ while staying $\geq p$ before.*

Proof. Suppose both fail, witnessed by arms e_1 (down to $p - 3$ at d_1) and e_2 (up to $p + 3$ at d_2). If $e_1 = e_2$, then whichever of d_1, d_2 comes first violates the bound in force before the other. Otherwise apply Lemma 6.1 with left arm e_1, right arm e_2, $\tau = 0$, $a = p + 1$ and $b = p - 1$. □

Lemma 6.4 (An up phase is not stuck; Lean: `nofail_up`)**.** *Assume $b \leq a$ and:*

- *every live arm e is $\geq b$ before its start time τ_e;*
- *some arm e_0 has a time d with $f_{e_0}(d) = a + 2$, $f_{e_0} \geq b$ before d, and $\tau_{e_0} \leq d$ if e_0 is live.*

Then ($\mathrm{N}_\uparrow$) *holds.*

Proof. Suppose a live arm e drops to $b - 2$ at $d' \geq \tau_e$ while staying $\leq a - 1$ on $[\tau_e, d')$. If $e = e_0$: we cannot have $d' < d$ (level $b-2 < b$) or $d' = d$. So $d < d'$ and $d \geq \tau_e$, which forces $f_e(d) \leq a-1 < a+2$. If $e \neq e_0$, the pair is the configuration of Lemma 6.1, with left arm e and right arm e_0. □

Lemma 6.5 (A down phase is not stuck; Lean: `nofail_down`)**.** *Consider a down phase from top a, following an up phase from bottom $b \leq a$. Every arm that is live in the down phase, with start time σ_e, was live in the up phase with start $\tau_e \leq \sigma_e$. It is $\geq b$ before τ_e and $\leq a - 2$ on $[\tau_e, \sigma_e)$. Assume some arm e_0, live in both phases, drops at some $d > \sigma_{e_0}$ to $b - 2$ while staying $\leq a - 1$ on $[\tau_{e_0}, d)$. Then no arm e live in the down phase rises to $a + 2$ at a time $rise \geq \sigma_e$ while staying $\geq b + 1$ on $[\sigma_e, rise)$.*

Proof. If $e = e_0$, compare *rise* with d as in the previous proof. If $e \neq e_0$, the pair is the configuration of Lemma 6.2, with left arm e and right arm e_0. □

7 The main induction and the end of the proof

Proposition 7.1 (Lean: `main_up`)**.** *Let $b+2 \le a \le H+1$ and $L-1 \le b$. Suppose the live arms have start times τ_e with $|f_e(\tau_e) - b| \le 1$, every live arm is $\ge b$ before τ_e, and* ($\mathrm{N}_\uparrow$) *holds. Then there is a list ts such that* $\mathrm{Fit}(f_e, \tau_e, b, ts)$ *holds for every live arm e.*

Proof. Strong induction on $N = (H+1-a) + (b-L+1) \ge 0$.

Up phase. Lemma 5.1 gives $t \ge a+1$ and new starts σ_e. If no σ_e is defined, then $ts = [t]$ works. Otherwise, Lemma 5.1(3) gives $t \le H+1$ and an arm e_0 that later drops to $b-2$ at $d > \sigma_{e_0}$ while staying in $[b-1, t-1]$. This is exactly the hypothesis of Lemma 6.5 with top t. So the down phase from top t is not stuck.

Down phase. The reflected form of Lemma 5.1 gives $t_2 \le b-1$ and new starts σ'_e. If none is defined, then $ts = [t, t_2]$ works. Otherwise $t_2 \ge L-1$, and some arm e_1 reaches t_2+1 at σ'_{e_1} and later rises to $t+2$ while staying in $[t_2+1, t+1]$.

Recursion. Take the new state: bottom t_2, previous top t, start times σ'_e. Its hypotheses hold:

- every live arm is $\ge t_2$ before its new start: before τ_e it is $\ge b > t_2$, on $[\tau_e, \sigma_e)$ it is $\ge b-1 \ge t_2$, and on $[\sigma_e, \sigma'_e)$ it is $\ge t_2+2$;
- ($\mathrm{N}_\uparrow$) holds by Lemma 6.4, with determinant arm e_1;
- the measure drops: $(H+1-t) + (t_2-L+1) < N$, because $t > a$ and $t_2 < b$.

The induction hypothesis gives *rest*, and $ts = [t, t_2, rest]$ works. (Each leg is fitted as explained after Lemma 5.1.) □

Proposition 7.2 (Lean: `greedy_fit`)**.** *There are T_0 with $|p - T_0| \le 1$ and a list ts such that both arms satisfy* $\mathrm{Fit}(f_e, 0, T_0, ts)$.

Proof. Use Lemma 6.3. In case (A), start an up phase from the pseudo-state with bottom $b = p-1$, previous top $a = p+1$, and both arms live at time 0. Its hypothesis ($\mathrm{N}_\uparrow$) is exactly (A), and Proposition 7.1 gives the list, with $T_0 = p-1$. In case (B), run one down phase from top $p+1$ (previous bottom $p-1$). Its "not stuck" hypothesis is exactly (B). Then continue with Proposition 7.1 as in its recursion step, with $T_0 = p+1$. □

Proof of Theorem 2.5. Proposition 7.2 and Lemma 4.3, applied to each arm with $\tau = 0$, give time assignments of both arms onto the *same* jump walk c of (T_0, ts). These are exactly folds in the sense of Definition 2.3. □

Proof of Theorem 1.4. If $\max \rho \le \min \rho + 2$, use the one-block consolidation. Otherwise Theorem 2.5 gives a common fold of the two arms of ρ from i. Proposition 2.4 turns it into the required block map. □

8 Correspondence with the Lean files

All file names are relative to the directory `HPC/`.

Here	Lean declaration	File
Def. 1.1, 1.2	`Chain`, `Chain.IsCrookedIn'`	`Chains/Basic.lean`
Def. 1.3	`IsBlockMap`, `Chain.ofBlocks`	`Chains/Consolidation.lean`
ρ, σ, Lemma 2.2	`Refines.minIdx`, `maxIdx`, `minIdx_lip`	`Chains/Consolidation.lean`
Theorem 1.4	`HPC.Fold.theorem3`	`Fold/Open.lean`
Reduction	`theorem3_of_S3`	`Fold/Span.lean`
Def. 2.3	`FoldsOnto`, `armG`, `armT`	`Fold/Itinerary.lean`
Prop. 2.4	`exists_blocks_first_of_folds`	`Fold/Itinerary.lean`
Degenerate case	`exists_blocks_first_of_range_le_two`	`Fold/Reduction.lean`
Faithful model	`IsCrookedIvF`, `isCrookedIvF_minIdx_maxIdx`	`Fold/Faithful.lean`
Theorem 2.5	`S3`, `Z77.s3_zig`, `s3_general`	`Fold/Span.lean`, `Fold/Zig77Main.lean`, `Fold/General.lean`
Lemma 3.1	`CrkOn`, `crkOn_of_isCrookedIvF`, `CrkOn.reverse`	`Fold/Zig77Crook.lean`
Section 4	`jwalk`, `jlen`, `Env`, `arr`, `Fit`, `fold_of_fit`	`Fold/Zig77Walk.lean`
Lemma 5.1	`up_step`, `down_step`	`Fold/Zig77Step.lean`
Lemmas 6.1, 6.2	`up_core`, `down_core`	`Fold/Zig77Crook.lean`
Lemmas 6.3–6.5	`first_direction`, `nofail_up`, `nofail_down`	`Fold/Zig77Main.lean`
Props. 7.1, 7.2	`main_up`, `greedy_fit`	`Fold/Zig77Main.lean`

Remarks on faithfulness.

- The formal theorem is the "block reading" of Bing's Theorem 3. The consolidation is described by a block map (a Lipschitz surjection of indices), and "lies in three consecutive links of D" is the three-link window $D_{[j,j+3)}$.
- The hypothesis $n > 0$ in the Lean statement is implied by the existence of $i \in [n]$.
- Most of the repository consists of earlier approaches to `S3`: other families of walks, span-by-span case analyses, and counterexample audits. The proof described here does not use them. Besides the basic chain library, it uses only the reduction and assembly lemmas of Section 2, the lemma `isCrookedIvF_minIdx_maxIdx`, and the four files `Fold/Zig77*.lean`.